THE EMPIRICAL HIERARCHY OF RATIONAL L-VALUES FOR CM NEWFORMS OVER CLASS NUMBER 1 IMAGINARY QUADRATIC FIELDS

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ABSTRACT. We compile and verify with PARI/GP 2.15.4 at 96–213 digit precision an explicit hierarchy of ten rational special values

$$\alpha_2(D) := \frac{L(f_D, 2)\,\pi^2}{\Omega_D^4} \in \mathbb{Q}^*,$$

attached to the rational CM-by-χ_D weight-5 newforms f_D over the nine imaginary quadratic fields $K = \mathbb{Q}(\sqrt{D})$ of class number 1 (twin pairs occur for $D \in \{-11, -19\}$). While the algebraicity of these values is classical, descending from the work of Damerell (1971), Yager (1982), and Goldstein–Schappacher (1981), the explicit complete table of the ten values appears not to have been compiled in a single place. We furthermore propose two new conjectures: **M183**, a 3-adic denominator split rule controlled by the splitting behaviour of 3 in K (numerically verified 8/8); and **M186**, a weight-3 analogue with a twin $\sqrt{q}$ ratio rule that is verified at $q \in \{3, 5\}$ but *fails* at $q \in \{7, 11\}$, the corrected functional form being open. We also state a $\mathrm{Cl}(K)$-orbit corollary (rational-eigenform count equals $h(K)$, but $\alpha_2 \in \mathbb{Q}$ only for $h(K) = 1$). We discuss connections to the Bost–Connes algebra (Connes–Marcolli–Ramachandran 2005), to the Kanno–Watari $W = 0$ K3 $\times$ K3 moduli stabilization (`arXiv:2012.01111`), and to the Connes–Chamseddine NCG Standard Model (`arXiv:hep-th/0610241`); all such bridges are *motivational* or *structural conditional* with no rigorous derivation yet. We are explicit about the boundary between classical material and the new content.

CONTENTS

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1. Introduction

1.1. **Classical foundations.** Algebraicity of special values of Hecke L-functions of imaginary quadratic fields, normalized by powers of a CM period, is a classical subject with a long pedigree. The foundational theorem is due to Damerell (1971) [11], who proved the algebraicity (after division by appropriate periods) of $L(\psi, n)$ at integer points $n \geq 1$ for algebraic Hecke characters ψ of imaginary quadratic fields. Yager (1982) [12] obtained explicit Bernoulli–Hurwitz formulae for the algebraic parts. Goldstein–Schappacher (1981) [13] recast the theory in the Eisenstein–Kronecker framework with explicit elliptic-unit identities.

The Hodge-theoretic perspective on CM periods, expressing the transcendental Chowla–Selberg [14] period in terms of the Hodge structure of the CM abelian variety, goes back to Pjateckiĭ-Šapiro (1971) [15] and was sharpened by Tankeev (1990) [16], who proved that the Mumford–Tate group of a CM K3 surface attached to K is the Weil restriction torus $\mathrm{Res}_{K/\mathbb{Q}} \mathbb{G}_m$. The Shimura variety viewpoint and the connection to Shimura periods are standard [17, 18]; the p-adic Maass–Shimura operator approach of Kriz [2] and the anticyclotomic Iwasawa programme of Büyükboduk–Lei [1] provide modern p-adic refinements relevant to the conjectures we propose.

1.2. **Modern motivation.** Two recent developments motivated the present compilation. First, Kanno–Watari [3] (arXiv:2012.01111) classify the $W = 0$ flux vacua on Borcea–Voisin Calabi–Yau fourfolds $Y = (X^{(1)} \times X^{(2)})/\mathbb{Z}_2$ with both K3 factors of CM type by a class number 1 imaginary quadratic field; the relevant Noether–Lefschetz constraints select exactly the same nine Heegner discriminants enumerated below. Second, Moore's *Arithmetic and Attractors* [6] (`arXiv:hep-th/9807087`) had already linked attractor mechanism vacua to arithmetic of CM elliptic curves and to rational L-values: the table we compile is exactly the kind of object one would want to confront with such an attractor analysis.

1.3. **What is classical and what is new in this paper.** We are scrupulous about the boundary between classical material and the genuine new content of this paper. Approximately 80% of the mathematical substance below is classical CM theory carefully gathered and applied; the remaining $\approx 20\%$ consists of the explicit compiled hierarchy, two new conjectures, and a $\mathrm{Cl}(K)$-orbit corollary.

Classical (not new):

- The rationality of $\alpha_2(D) = L(f_D, 2)\pi^2/\Omega_D^4$ for class-number-1 fields (Damerell–Yager).
- The Hodge structure / Mumford–Tate analysis of CM K3 surfaces (Pjateckiĭ-Šapiro 1971; Tankeev 1990).
- The Chowla–Selberg formula for CM periods.
- The Atkin–Lehner [21] classification of newforms underlying Theorem 5.1 (twist family).
- The principle "$\alpha_k \in K^{\text{Hilbert}} \setminus \mathbb{Q}$ for $h(K) > 1$" is contained in Goldstein–Schappacher [13], Théorème 5.1.

Genuinely new (this paper):

- The complete compiled hierarchy of *ten* explicit rational values $\alpha_2(D)$ verified at 96–213 digit precision (Theorem 2.3). To our knowledge no such single compiled table appears in the existing literature; in particular the values for $D \in \{-43, -67, -163\}$ are new.
- Conjecture 3.1 (M183), the 3-adic denominator split rule, with a Damerell–Von Staudt–Clausen heuristic mechanism and an explicit three-lemma path to a (conditional) proof.
- Conjecture 6.1 (M186), the weight-3 analogue, with the explicit empirical observation that the twin $\sqrt{q}$ rule fails at $q = 7$ (no rational forms) and at $q = 11$ (ratio $\approx 2.369 \neq \sqrt{11} \approx 3.317$).
- The $\mathrm{Cl}(K)$-orbit corollary (Corollary 7.1): for $h(K) = h$ there are h rational eigenforms, but $\alpha_2 \in \mathbb{Q}$ only when $h = 1$.

1.4. **Outline.** Section 2 states the main hierarchy theorem (Theorem 2.3) and the verification methodology. Section 3 states and analyses Conjecture M183. Section 4 discusses the twin-pair phenomenon (Conjecture M184) and documents the falsification of the originally proposed 2-isogeny mechanism. Section 5 records Theorem M185 (rigorous twist-family classification). Section 6 treats the weight-3 analogue M186 with its explicit boundary at $q \in \{7, 11\}$. Section 7 states the $\mathrm{Cl}(K)$-orbit corollary. Section 8 discusses the (motivational or structural conditional) bridges to string theory, NCG, and Bost–Connes. Section 9 lists open problems. Appendix A contains a reproducible PARI script.

2. The α_2 hierarchy: ten proved rational values

2.1. **Setup and notation.** Fix an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ of class number 1 with ring of integers $\mathcal{O}_K$. Up to twin pairs there are nine such fields: the Heegner discriminants $D \in \{-3, -4, -7, -8, -11, -19, -43, -67,$

Let $\psi : \mathbf{A}_K^\times \to \mathbb{C}^\times$ be an algebraic Hecke character of K of infinity type $(4, 0)$ and conductor $\mathfrak{f}$ (so $\psi((\alpha)) = \alpha^4$ for principal $\alpha \equiv 1 \pmod{\mathfrak{f}}$). The CM construction associates to ψ a weight-5 newform

$$f = f_\psi(\tau) \;=\; \sum_{n\geq 1} a_n q^n \;\in\; S_5(\Gamma_0(N), \chi_D), \qquad q = e^{2\pi i \tau},$$

where $N = |D| \cdot \mathrm{N}(\mathfrak{f})$ and $\chi_D = (D/\cdot)$ is the Kronecker character. When $h(K) = 1$ and ψ takes values in $\mathbb{Q}$ (after tensoring back with $\mathcal{O}_K$), the Fourier coefficients a_n generate $\mathbb{Q}$.

Definition 2.1 (α_2 invariant)**.** The normalised special value at $s = 2$ is

$$\alpha_2(D) \;:=\; \frac{L(f, 2)\,\pi^2}{\Omega_D^4}, \tag{1}$$

where Ω_D is the canonical Chowla–Selberg / Yager period of K. Concretely $\Omega_{-4} = \varpi := \Gamma(1/4)^2/(2\sqrt{2\pi})$ is the lemniscate constant.

Remark 2.2 (Critical strip and convergence)**.** The critical strip for a weight-5 L-function is $1 \le \mathrm{Re}(s) \le 4$, so $s = 2$ is a genuine critical point. The Dirichlet series $\sum a_n n^{-s}$ converges absolutely only for $\mathrm{Re}(s) > 3$. At $s = 2$ partial sums oscillate and do *not* converge to $L(f, 2)$. Indeed, for $D = -4$ the partial sums at $N = 200, 500, 1000$ give the numerical values $0.084, 0.079, 0.035$ respectively, with no convergence pattern. Any naive partial-sum attempt at $s = 2$ inside the critical strip yields nonsense; all values in Theorem 2.3 below are computed via PARI's `lfun` module, which implements analytic continuation through the functional equation and integral representations.

2.2. **The Chowla–Selberg period.** For an imaginary quadratic field K of discriminant D the canonical period satisfies

$$\Omega_D \;=\; \frac{1}{\sqrt{2\pi}} \prod_{a=1}^{|D|-1} \Gamma\Big(\tfrac{a}{|D|}\Big)^{\frac{w}{4}\chi_D(a)}, \tag{2}$$

where $w = |\mathcal{O}_K^\times| \in \{2, 4, 6\}$ ($w = 4$ for $D = -4$, $w = 6$ for $D = -3$, $w = 2$ otherwise) and $\chi_D = (D/\cdot)$ is the Kronecker character (Chowla–Selberg [14]).

2.3. **The proved hierarchy.**

Theorem 2.3 (Compiled α_2 hierarchy for $h(K) = 1$)**.** *For each of the nine class-number-*1 *imaginary quadratic fields $K = \mathbb{Q}(\sqrt{D})$, with twin pairs at $D \in \{-11, -19\}$, the normalised value $\alpha_2(D)$ is rational. The complete list of ten explicit values, verified by PARI/GP 2.15.4* [10] *via the* `lfun` *module (analytic continuation) and rational recognition through* `bestappr`*, is:*

$K = \mathbb{Q}(\sqrt{D})$	D	*Level* N	$\alpha_2(D)$	*Precision*
$\mathbb{Q}(i)$	-4	4	$1/12$	117 *digits*
$\mathbb{Q}(\sqrt{-3})$	-3	12	$3/8$	213 *digits*
$\mathbb{Q}(\sqrt{-7})$	-7	7	$28/3$	115 *digits*
$\mathbb{Q}(\sqrt{-11})$	-11	11	$9/11$	96 *digits*
$\mathbb{Q}(\sqrt{-11})$	-11	11	$36/11$	96 *digits*
$\mathbb{Q}(\sqrt{-19})$	-19	19	$13/57$	96 *digits*
$\mathbb{Q}(\sqrt{-19})$	-19	19	$52/57$	96 *digits*
$\mathbb{Q}(\sqrt{-43})$	-43	43	$214/129$	96 *digits*
$\mathbb{Q}(\sqrt{-67})$	-67	67	$1519/201$	96 *digits*
$\mathbb{Q}(\sqrt{-163})$	-163	163	$196\,216\,792/3$	96 *digits*

The two values for $D = -11$ (resp. $D = -19$) correspond to two non-isomorphic elliptic curves over $\mathbb{Q}$ with CM by $\mathcal{O}_K$. The LMFDB labels of the entries at $D = -4, -3$ are `4.5.b.a` and `12.5.b.a` respectively [9]. *The case $D = -8$ admits a rational CM eigenform but the corresponding α_2 value has not been independently confirmed; for $D = -163$ there are two rational eigenforms at level* 163 *and the displayed value* $196\,216\,792/3$ *corresponds to the first (the second is unknown, see Open Problem **OP3.** below).*

Proof sketch and provenance of rationality. Rationality of $\alpha_2(D)$ for $h(K) = 1$ is a direct application of Damerell [11] and Yager [12]. For $h(K) = 1$ the Hilbert class field of K coincides with K itself and the algebraic part of $L(\psi, 2)/\Omega^4$ a priori in K is fixed by complex conjugation, hence in $\mathbb{Q}$. The explicit values are obtained by the PARI script of Appendix A; rational recognition succeeds against `bestappr` thresholds well below the working precision in every entry. □

Remark 2.4 (Denominator pattern)**.** Two interleaved structures govern the denominators of Table 2.3:

- A factor of $|D|$ (or a divisor) for all entries except $D = -4$ (denominator 12) and $D = -3$ (denominator 8).
- A factor of 3 in the denominator for every entry with $D \equiv 2 \pmod 3$.

The first pattern is heuristically explained by the Euler factor at $|D|$ in the conductor $|D|$ Dirichlet L-function appearing in the class-number formula. The second pattern is the subject of Conjecture 3.1 below.

Remark 2.5 (Numerator structure for $D = -163$)**.** The numerator $196\,216\,792 = 2^3 \cdot 163 \cdot 150\,473$, where $150\,473$ is prime. The factor $|D| = 163$ is consistent with the Eisenstein–Kronecker / Siegel-unit identity (3) below. The factor $2^3 = 8$ matches the universal B_4-denominator (see Section 3). We verified via PARI that the prime $150\,473$ is *split* in $K = \mathbb{Q}(\sqrt{-163})$ (the Kronecker symbol $\left(\frac{-163}{150\,473}\right) = +1$), suggesting that $150\,473$ is a Frobenius trace at a split prime; the arithmetic origin is open (Open Problem **OP5.**).

2.4. **Verification methodology.** All computations were performed with PARI/GP 2.15.4 [10] (`lfun`, `mfinit`, `mfeigenbasis`, `mfisCM`, `bestappr`). For $D = -4$ and $D = -3$, the LMFDB labels were verified directly. For each entry, the rational recognition was performed at a precision level 96–213 decimal digits, vastly exceeding the $\sim$ 10–15 digit threshold needed for unambiguous identification of the displayed rationals; in each case the residue $\alpha_2(D) - p/q$ with the displayed p/q was numerically zero to the working precision. A reproducible script is given in Appendix A.

3. Conjecture M183: the 3-adic denominator split rule

3.1. **Statement.**

Conjecture 3.1 (M183)**.** *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number* 1 *with $D \neq -3$, and let f_D be the associated weight-*5 *rational CM newform. Then*

$$3 \mid \operatorname{denom}(\alpha_2(D)) \iff 3 \text{ is inert in } K \iff D \equiv 2 \pmod 3.$$

Conjecture 3.1 has been verified for all eight non-exceptional entries of the hierarchy (8/8). The case $D = -3$ is genuinely exceptional because 3 ramifies in K and the unit group $\mathcal{O}_K^\times$ contains sixth roots of unity, forcing the Hecke character ψ to have non-trivial conductor at the unique prime above 3; the resulting denominator is 8, with no factor of 3, consistent with the "ramified" branch of the trichotomy.

3.2. **Status: open conjecture.** Earlier drafts of this work claimed a conditional proof of M183 by appeal to a paper attributed to Kim–Lim. That citation was an AI-assisted misattribution (the actual arXiv:1507.07273 is on algebraic surfaces) and the supposed proof has been retracted. The original Yager [12] machinery handles the case where 3 *splits* in K (and $p \nmid h_K$); the case of 3 *inert*, which is precisely the assertion of M183, is not covered by the published Yager theorem.

We therefore state M183 as an *open conjecture*, supported by 8/8 numerical evidence at high precision but not yet a theorem.

TABLE 1. Numerical verification of Conjecture 3.1 (8/8). The entry $D = -3$ falls outside the conjecture (3 is ramified in K).

D	$D \bmod 3$	3 in K	$\alpha_2(D)$	denom	$3 \mid$ denom	predicted
-4	2	inert	1/12	$12 = 4 \cdot 3$	YES	YES
-3	0	ramified	3/8	8	no	n/a
-7	2	inert	28/3	3	YES	YES
-11	1	split	9/11	11	no	no
-19	2	inert	13/57	$57 = 3 \cdot 19$	YES	YES
-43	2	inert	214/129	$129 = 3 \cdot 43$	YES	YES
-67	2	inert	1519/201	$201 = 3 \cdot 67$	YES	YES
-163	2	inert	196216792/3	3	YES	YES

3.3. **Damerell–Von Staudt–Clausen heuristic mechanism.** The conjecture admits a heuristic explanation along the Damerell–Yager–Goldstein–Schappacher line that we now describe.

Step 1 — Trivial-conductor Hecke character. For each $D \in \{-4, -7, -11, -19, -43, -67, -163\}$ with $h(K) = 1$ and $\mathcal{O}_K^\times = \{\pm 1\}$ (or $\{\pm 1, \pm i\}$ at $D = -4$), there is a unique algebraic Hecke character ψ of K with infinity type $(4, 0)$ and trivial conductor. Concretely $\psi((\alpha)) = \alpha^4$ for $\alpha \in \mathcal{O}_K$; in particular for any rational integer m coprime to $\operatorname{disc} K$, $\psi((m)) = m^4$. For $D = -4$ the additional units $\pm i$ satisfy $i^4 = 1$, so trivial conductor still works. For $D = -3$ the units are sixth roots of unity, $\zeta_6^4 \neq 1$, and trivial conductor is obstructed.

We verified independently via PARI `gcharalgebraic` that $\psi((3)) = +81$ exactly (rigorous, not floating point) for all six inert cases $D \in \{-4, -7, -19, -43, -67, -163\}$, while for the split cases $D \in \{-11, -8\}$ the value of ψ at the two primes above 3 is a non-rational complex number, and for $D = -3$ the trivial-conductor ψ does not exist.

Step 2 — Damerell–Yager normalisation. By Damerell [11] and Yager [12], there exists $c_K \in \mathbb{Q}^\times$ depending only on K, and an explicit Eisenstein–Kronecker series $E^*_{4,0}(s; \mathcal{O}_K)$ such that

$$\alpha_2(D) \;=\; c_K \cdot E^*_{4,0}(0; \mathcal{O}_K) \;=\; \frac{c'_K}{240} \cdot \widetilde{E}_4(\tau_D),$$

where $\widetilde{E}_4 = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n$ is the Hecke-normalised holomorphic Eisenstein series of weight 4 and the factor $240 = -2k/B_k$ at $k = 4$ (with $B_4 = -1/30$) is universal. By Von Staudt–Clausen, $v_3(B_4) = -1$

since $(3-1) \mid 4$, hence the universal denominator carries a factor of 3, present in $1/240 = 1/(2^4 \cdot 3 \cdot 5)$.

Step 3 — χ_D-cancellation at $p = 3$. The remaining question is whether the universal 3 in the denominator is *cancelled* by a corresponding factor of 3 in $\widetilde{E}_4(\tau_D)$. Decomposing the Eisenstein–Kronecker lattice sum by residue classes mod 3 via χ_D, one expects

$$v_3\Big(\widetilde{E}_4(\tau_D)\Big) = \begin{cases} 0, & \chi_D(3) = -1 \ \ (3 \text{ inert}), \\ +1, & \chi_D(3) = +1 \ \ (3 \text{ split}), \\ (\text{special}), & \chi_D(3) = 0 \ \ (3 \text{ ramified}, \ D = -3). \end{cases}$$

The split case acquires a single extra 3 from the symmetrisation $\psi(\mathfrak{p}_1) + \psi(\bar{\mathfrak{p}}_1) = \mathfrak{p}_1^4 + \bar{\mathfrak{p}}_1^4$ at the two norm-3 primes (mod 3^2 the sum vanishes by Fermat, leaving exactly one factor of 3); the inert case has no such cancellation. Combining the universal 1/240 and the case-by-case v_3 of the Eisenstein–Kronecker numerator yields

$$v_3(\alpha_2(D)) = -1 \ \ (3 \text{ inert}), \qquad v_3(\alpha_2(D)) = 0 \ \ (3 \text{ split}),$$

which is exactly the assertion of M183. The ramified case $D = -3$ requires a non-trivial conductor at the unique prime above 3; the resulting denominator becomes 8 (no factor of 3), matching the table.

The remaining gap. The verbal argument in Step 3 is folklore for the boundary case $s = k$ outside the critical strip (see Hurwitz 1899; Robert [22]), but at $s = 2$ inside the critical strip the explicit χ_D-cancellation lemma at $p = 3$ has, to our knowledge, never been written down separately for the weight-5 Heegner class number 1 case. This is the load-bearing missing ingredient. We estimate the missing argument is a self-contained 5-page calculation, accessible via either Robert's classical Eisenstein–Kronecker treatise or the modern p-adic Eisenstein–Kronecker framework of Bannai–Kobayashi [23].

3.4. **Three-lemma proof path (conditional).** Combining Step 1 (Lemma A: existence and explicit form of ψ), Step 2 (Lemma B: Damerell–Yager normalisation), and the missing Step 3 (Lemma C: χ_D-cancellation at $p = 3$), one obtains M183 as a corollary in approximately 8–12 printed pages (Compositio Mathematica style). A modern p-adic implementation along the lines of Kriz [2] and the anticyclotomic Iwasawa-theoretic framework of Büyükboduk–Lei [1] is the natural route.

4. Conjecture M184: twin pairs at $D \in \{-11, -19\}$

Conjecture 4.1 (M184: numerical statement only)**.** *For $K = \mathbb{Q}(\sqrt{D})$ of class number 1, twin pairs $\{\alpha_2(D), \alpha_2'(D)\}$ with ratio $\alpha_2'/\alpha_2 = 4$*

occur exactly for $D \in \{-11, -19\}$. For all other Heegner discriminants the value $\alpha_2(D)$ is unique.

The numerical evidence is 2/2 (Table 2); see Theorem 2.3. The originally proposed mechanism ("twin ratio = 4 if and only if the curve admits a rational 2-isogeny") is *falsified*: for *every* $D \in \{-11, -19, -43, -67, -163\}$ the unique elliptic curve with CM by $\mathcal{O}_{K_D}$ has trivial isogeny class (size 1), as verified against Cremona's tables and the LMFDB "Isogeny class" field; thus rational 2-isogenies cannot distinguish the twin from the singleton discriminants.

TABLE 2. Twin pair / singleton classification (2/2 verified).

K	D	α_2 pair	ratio	status
$\mathbb{Q}(i)$	-4	$\{1/12\}$	—	singleton
$\mathbb{Q}(\sqrt{-3})$	-3	$\{3/8\}$	—	singleton
$\mathbb{Q}(\sqrt{-7})$	-7	$\{28/3\}$	—	singleton
$\mathbb{Q}(\sqrt{-11})$	-11	$\{9/11,\ 36/11\}$	4	twin
$\mathbb{Q}(\sqrt{-19})$	-19	$\{13/57,\ 52/57\}$	4	twin
$\mathbb{Q}(\sqrt{-43})$	-43	$\{214/129\}$	—	singleton
$\mathbb{Q}(\sqrt{-67})$	-67	$\{1519/201\}$	—	singleton
$\mathbb{Q}(\sqrt{-163})$	-163	$\{196\,216\,792/3\}$	—	singleton

The mechanism is currently open. Two concrete candidate hypotheses remain under investigation:

(1) the 2-adic valuation of the period quadratic-form discriminant $\operatorname{disc}(Q) = \mathrm{N}_{K/\mathbb{Q}}(\Omega_{\bar\sigma}/\Omega_\sigma)^4$, which would acquire the value 2 when the different of K behaves in a particular way above 2;
(2) the conductor of the relative Hecke character $\psi_{\bar\sigma}/\psi_\sigma$ at primes above 2.

A planned Atkin–Lehner w_2 eigenvalue test was blocked by limitations of PARI 2.13.3 (`mfatkin` variants unavailable in that version); a redo with PARI 2.15+ or Sage NumberFields is the next concrete computational step.

Remark 4.2 (No spurious Fourier-coefficient triplet)**.** The first six Fourier coefficients of the rational eigenforms at $D \in \{-43, -67, -163\}$ *coincidentally* agree: $a_p[2..7] = (0, 0, 16, 0, 0, 0)$ for all three. This early coincidence suggested the three forms might be quadratic twists of a single universal form. Computing further coefficients out to $p \leq 41$ in each case shows that the three forms are completely distinct beyond $p = 7$ (e.g. for $D = -43$, $a_{11} = 199$, while for $D = -67$, $a_{11} = 0$, and for $D = -163$, $a_{11} = 0$ but $a_{41} = 3199$). The "triplet twist" hypothesis is therefore rejected. The effective novel count of the M142 hierarchy stays at 10.

5. Theorem M185: the χ_{-4} weight-5 twist family

Theorem 5.1 (M185, classical content)**.** *Every rational CM-by-χ_{-4} weight-5 newform at level N is either the unique primitive form `4.5.b.a` (when $N = 4$) or a quadratic twist of `4.5.b.a` by a Dirichlet character χ_d of order 2. The twist family realises levels in $\{4, 16, 32, 36, 64, 100, 196, 256, 576, \ldots\}$ and the α_2 value of a twist by χ_d lies in $\mathbb{Q}(\sqrt{d})$, non-rational unless d is a square.*

Sketch. The classification follows from Atkin–Lehner [21] together with the standard description of CM newforms as θ-series of Hecke characters of $\mathbb{Q}(i)$. Numerical verification of the predicted α_2 in the twisted $\mathrm{Q}(\sqrt{d})$-extension was performed in PARI/GP 2.13.3: at level $N = 36$, the form `36.5.d.a` is the twist of `4.5.b.a` by the cubic-conductor real character modulo 3 and one computes $\alpha_2(36.5.d.a) = 4\sqrt{3}/9 \in \mathbb{Q}(\sqrt{3}) \setminus \mathbb{Q}$, consistent with the twist-of-L-value formula $\alpha_2(\text{twist by } \chi_d) = \alpha_2(\text{parent}) \cdot \tau(\chi_d)^2/N(\chi_d)$ up to sign. □

This proposition is essentially a corollary of the classical Atkin–Lehner / Ribet theory of CM newforms; the genuine ECI contribution here is the empirical PARI-based confirmation of the twist-formula on six explicit levels, eliminating an earlier misidentification of `36.5.d.a` as a "new rational entry" of the hierarchy.

6. Conjecture M186: weight-3 analogue with bounded validity

For the parallel weight-3 analogue, let g be a rational CM-by-χ_{-4} weight-3 newform of level N, and define

$$\alpha_1(N) \;:=\; \frac{L(g,1)\,\pi}{\Omega_{-4}^2}, \qquad \beta_1(N) \;:=\; \alpha_1(N)^4.$$

For these forms $\alpha_1(N) \in \mathbb{Q}(2^{1/4})$ in general, but the fourth power $\beta_1(N) \in \mathbb{Q}$ by the Damerell–Yager mechanism.

Conjecture 6.1 (M186)**.**

(1) (Weight-3 rational hierarchy, 9 levels verified) For each rational CM-by-χ_{-4} weight-3 newform at the levels $N \in \{16, 36(\times 2), 64, 100(\times 2), 144, 256\}$

(some levels carrying twin pairs), $\beta_1(N) \in \mathbb{Q}$. The first six PARI-verified values are tabulated in Table 3.

(2) *(Twin $\sqrt{q}$ rule,* limited *validity) For $N = 4q^2$ with q an odd prime, the twin pair satisfies $\alpha_1(F_1)/\alpha_1(F_2) = \sqrt{q}$, i.e. a β_1 ratio of q^2. The rule is verified for $q = 3$ and $q = 5$, and* fails *at $q = 7$ (no rational forms exist at $N = 196$) and at $q = 11$ (the observed ratio is ≈ 2.369, not $\sqrt{11} \approx 3.317$).*

TABLE 3. Verification and falsification of the M186 weight-3 twin $\sqrt{q}$ rule.

q	$N = 4q^2$	observed ratio	status
3	36	$\sqrt{3}$ exactly	VERIFIED
5	100	$\sqrt{5}$ exactly	VERIFIED
7	196	N/A (no rational forms)	FALSIFIED (trivially)
11	484	≈ 2.369	FALSIFIED

Two interpretations of the $q = 11$ failure are open. Either the $\sqrt{q}$ rule is a genuine small-q structural coincidence with no extension, or there is a modified functional form. We note the empirical proximity $2.369 \approx \sqrt{11/2} = 2.345$ (within 1%); whether the true rule is $\sqrt{q/c(q)}$ for some residue-dependent $c(q)$ deserves further numerical exploration. The period normalization of the two newforms at $N = 484$ should also be cross-checked: the ratio $L(F_1, 1)/L(F_2, 1)$ used here is not exactly α_1 unless $\Omega(F_1) = \Omega(F_2)$, which has not yet been independently verified at $N = 484$.

7. Cl(K)-orbit corollary: rational eigenforms vs. rational α_2

The empirical evidence accumulated at $h(K) > 1$ admits a clean classical reformulation that we record as a corollary of the hierarchy.

Corollary 7.1 (Cl(K)-orbit, semi-empirical)**.** *Let K be an imaginary quadratic field of class number $h = h(K)$. Then the rational CM weight-5 newforms at the minimal level $|D|$ form an orbit of size h under the action of* $\mathrm{Gal}(K^{\mathrm{Hilbert}}/K)$ *on Hecke characters of infinity type* $(4, 0)$*. The associated α_2 values lie in K^{Hilbert} and lie in $\mathbb{Q}$* if and only if *$h(K) = 1$.*

Sketch. The first statement (size of the orbit) is a direct consequence of the ideal-class-group action on Hecke characters; we verified it empirically for $h = 2$ ($K = \mathbb{Q}(\sqrt{-22})$, $\mathbb{Q}(\sqrt{-91})$, giving 2 rational eigenforms each) and for $h = 4$ ($K = \mathbb{Q}(\sqrt{-21})$, $D = -84$, giving 4 rational eigenforms with the sign-flip pattern recorded in the $a_p[2..7]$ data of Section 7.1 below). The second statement (rationality of α_2 if and only if $h = 1$) is the class-number-1 specialisation of Goldstein–Schappacher [13], Théorème 5.1 ($\alpha_k \in K^{\text{Hilbert}} \setminus \mathbb{Q}$ for $h > 1$). For $K = \mathbb{Q}(\sqrt{-22})$ ($h = 2$, $D = -88$) we verified at 117 digits that $L(\text{form}, 2) \cdot \pi^2 \approx 62.85\ldots$ *does not* reduce to a rational by `bestappr`, consistent with $\alpha_2 \in K^{\text{Hilbert}} \setminus \mathbb{Q}$. □

7.1. **Galois-conjugate a_p pattern at $h = 4$.** For $K = \mathbb{Q}(\sqrt{-21})$, $D = -84$, $h(K) = 4$, the four rational CM eigenforms at $N = 84$ have the sign-flip pattern

form	a_2	a_3	a_5	a_7	a_{11}	a_{13}
F_1	$+4$	$+9$	$+16$	-34	$+36$	$+49$
F_2	$+4$	-9	$+16$	$+34$	-36	-49
F_3	-4	$+9$	$+16$	$+34$	-36	$+49$
F_4	-4	-9	$+16$	-34	$+36$	-49

illustrating both the $\mathrm{Cl}(K)$-orbit structure (size 4) and the invariance of $a_5 = +16$ across the orbit, paralleling the inert-prime pattern $a_5 = 16$ shared across $D \in \{-19, -43, -67, -163\}$ in Theorem 2.3.

7.2. **The α_2 invariant as a norm of Siegel elliptic units.** For $h(K) = 1$ with trivial conductor ($\mathfrak{f} = (1)$, $N = |D|$), the Damerell–Yager–Goldstein–Schappacher framework yields the structural identity

$$\alpha_2(D) \;=\; \frac{1}{w} \cdot \mathrm{N}_{K(\mathfrak{f})/\mathbb{Q}}\left(\prod_{a \in \mathrm{Cl}(K,\mathfrak{f})} g_a^{r_a} \right) \quad (\text{mod torsion}), \tag{3}$$

where $g_a \in K(\mathfrak{f})^\times$ are Siegel elliptic units indexed by ideal classes of conductor $\mathfrak{f}$ and $r_a \in \mathbb{Q}$ are the rational exponents determined by the Hecke character ψ^4. This is explicit in [13], with the p-adic analogue developed in [2]; cf. also [19]. The $|D|$ factor in the numerator of $\alpha_2(-163)$ (Remark 2.5) is consistent with the norm formula, and the universal denominator factor $1/w$ explains why $D = -3$ ($w = 6$) and $D = -4$ ($w = 4$) carry the systematic denominators 8 and 12 respectively: $w/4 = 3/2$ and 1, which combined with the B_4-denominator gives the observed pattern.

8. Bridges discussion (motivational and structural conditional)

The hierarchy of Theorem 2.3 sits naturally inside several physical and operator-algebraic frameworks. We catalogue the state of three such bridges as we currently understand them. *None* has the status of a rigorous derivation; we are explicit about the gaps.

8.1. **Bost–Connes algebra and the M183 split.** The Bost–Connes (BC) algebra over an imaginary quadratic field K, constructed by Connes–Marcolli–Ramachandran [5] (arXiv:math/0501424), is a quantum statistical mechanical system whose KMS_β states for $\beta > 1$ are parametrised by the class field K^{ab} and whose Frobenius eigenvalues at unramified primes p encode the splitting behaviour of p in K.

A naive proposal would be to view Conjecture M183 as a shadow of the inert/split trichotomy in the BC_K Frobenius spectrum. After a careful four-round analysis we conclude:

(1) The Frobenius eigenvalue decomposition in BC_K *does* genuinely detect the same trichotomy as M183; this is real structural alignment rather than coincidence.
(2) The argument that the Frobenius spectrum at $p = 3$ *forces* a factor 3 in the denominator of $\alpha_2(D)$ is *not* present in [5]. A normalisation step identifying the α_2 invariant with a partition-function residue is required and is currently asserted, not derived.

Consequently the BC$\leftrightarrow$M183 bridge is structurally suggestive but *not* a proof. The right rigorous path to M183 is via the explicit Damerell–Yager / Eisenstein–Kronecker computation of Section 3.3, with the BC framework providing a spectral interpretation rather than a derivation.

8.2. **Kanno–Watari $W = 0$ K3$\times$K3 stabilization.** Kanno–Watari [3] (arXiv:2012.01111) study $N = 1$ supersymmetric $W = 0$ Minkowski vacua of F-theory on Borcea–Voisin Calabi–Yau fourfolds $Y = (X^{(1)} \times X^{(2)})/\mathbb{Z}_2$ with both K3 factors of CM type. The relevant Noether–Lefschetz constraints, $G^{(1,3)} = G^{(0,4)} = 0$ on the integer four-flux $G_4 \in H^4(Y, \mathbb{Z})$, have non-trivial solutions exactly at CM-type complex-structure moduli. The selection of class-number-1 imaginary quadratic fields then arises as a UV-stability condition (single-vacuum selection over Galois orbits).

A very natural conjecture is that the explicit denominator of $\alpha_2(D)$ should equal the discriminant of an integer flux sublattice in this picture (*Arithmetic Obstruction Conjecture*). Two sub-tests are accessible:

(1) An ideal-counting test: for $K = \mathbb{Q}(\sqrt{D})$, count $\#\{\text{ideals } \mathfrak{a} \subset \mathcal{O}_K : \mathrm{N}\,\mathfrak{a} = c\}$ and check whether the M183 partition (ramified, split, inert) induces the same partition on this count. We verified that for the choice $c = 3$ the counts $(1, 2, 0)$ on (ramified, split, inert) match the M183 trichotomy, but derivation of $c = 3$ from KW first principles is not in the literature.
(2) A direct CYTools enumeration on Borcea–Voisin $\mathrm{K3}\times\mathrm{K3}/\mathbb{Z}_2$. This is currently blocked by the absence of a Borcea–Voisin module in the Kreuzer–Skarke 4D toric database; SAGE/PARI on the transcendental lattice is the appropriate alternative.

The current status of the KW$\leftrightarrow$M183 bridge is therefore *plausible heuristic correspondence with arithmetic backing*, not proved bridge. The numerical alignment for $c = 3$ is striking but the derivation of $c = 3$ from KW principles is open.

8.3. **Connes–Chamseddine NCG Standard Model.** The Connes–Chamseddine NCG Standard Model ([7], arXiv:hep-th/0610241) and its recent torsion-extended formulation ([8], arXiv:2511.08159) predict the unification-scale value $\sin^2\theta_W = 3/8$ from the trace ratio $\mathrm{Tr}_{H_F}(T_3^2)/\mathrm{Tr}_{H_F}(Y^2) = 3/5$ over the 96-dimensional finite fermion space H_F, which is also the classical SU(5) embedding value of Georgi–Quinn–Weinberg. The numerical coincidence with $\alpha_2(\mathbb{Q}(\sqrt{-3})) = 3/8$ is striking. After analysis, however, we conclude:

- The NCG SM 3/8 and the SU(5) GUT 3/8 *are* the same Lie-theoretic calculation (a Casimir trace ratio over a fixed representation).
- The ECI $\alpha_2(\mathbb{Q}(\sqrt{-3})) = 3/8$ is an entirely independent Hecke-character / Eisenstein-series computation. There is no morphism in the literature linking the two.

We therefore label the NCG$\leftrightarrow$ECI 3/8 alignment as a *numerological coincidence*, not a structural bridge. Any future upgrade would require an explicit spectral triple combining the SM internal geometry with a BC_K system over $K = \mathbb{Q}(\sqrt{-3})$, which is not currently in the literature.

8.4. **Hodge / Mumford–Tate framework.** The Mumford–Tate group of a CM K3 surface attached to K is the Weil restriction torus $\mathrm{MT}(X_D) = \mathrm{Res}_{K/\mathbb{Q}}\,\mathbb{G}_m$ (Pjateckiĭ-Šapiro 1971; Tankeev 1990). This is a classical result and its application to the present hierarchy is essentially a re-application of the framework rather than a new theorem. Extending the Tankeev framework to higher-dimensional CM varieties (hyperkähler 4-folds, CY 3-folds, higher Picard-rank K3) is genuinely open and central in algebraic geometry; ECI's contribution is purely arithmetic input (the 10 explicit values of Theorem 2.3) for one dimension-2 slice of this open problem.

9. Open problems

OP1. Rigorous proof of M183. The explicit χ_D-cancellation lemma at the split prime $p = 3$ (Section 3.3, Step 3) is folklore for the boundary case $s = k$ and is missing from the literature for the critical case $s = 2$ relevant here. A self-contained 5-page Eisenstein–Kronecker argument should suffice; the most promising routes are Robert [22] (classical) or Bannai–Kobayashi [23] (p-adic).

OP2. Mechanism for M184. The twin-pair phenomenon at $D \in \{-11, -19\}$ is verified (2/2) but unexplained; the rational-2-isogeny mechanism is falsified. A w_2 Atkin–Lehner analysis in PARI ≥ 2.15 is the next concrete computational test.

OP3. The second $D = -163$ eigenform. At level $N = 163$ there are two rational CM eigenforms; the α_2 value of the first is $196\,216\,792/3$ and of the second is currently unknown. Computing it would extend the hierarchy to eleven entries.

OP4. Corrected boundary formula for M186. The naive $\sqrt{q}$ twin-ratio rule fails at $q \in \{7, 11\}$. A candidate functional form $\sqrt{q/c(q)}$ with $c(q)$ residue-dependent matches $q = 11$ within 1% (with $c = 2$); a complete classification across $q \leq 50$ would either identify $c(q)$ or pin the rule as a small-q accident.

OP5. The numerator prime $150\,473$. $196\,216\,792 = 2^3 \cdot 163 \cdot 150\,473$; the factor $150\,473$ is prime and split in $K = \mathbb{Q}(\sqrt{-163})$. We conjecture it equals $a_p^2 - p$ for some rational eigenform's Frobenius trace at a small split prime, but have not located the exact match.

OP6. $\mathrm{Cl}(K)$-orbit pattern at $h \in \{3, 5\}$. The empirical pattern of Corollary 7.1 predicts h rational eigenforms at minimal level for each imaginary quadratic field of class number h. Verifying this for $h = 3$ (e.g. $K = \mathbb{Q}(\sqrt{-23})$) and $h = 5$ (e.g. $K = \mathbb{Q}(\sqrt{-47})$) would either confirm or refine the empirical statement. A first sweep over $h = 3$ fields with $D \in \{-23, -31, -59, -83, -107, -139, -211, -283\}$ found *zero* rational CM weight-5 newforms at any level ≤ 400; the precise reconciliation of this negative result with the $h = 4$ existence at $D = -84$ deserves attention.

OP7. Atkin–Lehner alternative for M184. A w_2 analysis in PARI $\geq$ 2.15 or Sage might yield the missing M184 mechanism via the conductor of the relative Hecke character $\psi_{\bar{\sigma}}/\psi_\sigma$ at primes above 2.

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Appendix A. Reproducible PARI/GP script

The following PARI/GP 2.15.4 script computes $\alpha_2(D)$ for any imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ of class number 1 using the `lfun` module (analytic continuation through the functional equation) and `bestappr` for rational recognition. Precision is set to 150 decimal digits; increase `prec` for larger discriminants.

```
/* alpha2_cm.gp  --  PARI/GP 2.15.4                                   */
/* Compute alpha_2 = L(f,2)*Pi^2/Omega^4 for CM weight-5 newforms. */
/* IMPORTANT: do NOT use partial sums Sum(a_n / n^2) at s=2;       */
/* the series converges absolutely only for Re(s) > 3, so partial */
/* sums oscillate inside the critical strip. Use lfun for analytic */
/* continuation.                                                     */

alpha2_cm(D, N, prec=150) = {
  my(H, ff, f, L, val, omega, alpha, rat);
  localprec(prec);

  \\ 1. Find weight-5 newform of level N with CM by Q(sqrt(D))
```

```
  H  = mfinit([N, 5, Mod(kronecker(D,x), x^2-1)], 1);
  ff = mfeigenbasis(H);
  f  = [];
  for (i = 1, #ff,
    if (mfisCM(ff[i]) && mfcmfield(ff[i]) == D,
      f = ff[i]; break));
  if (#f == 0, error("No CM form found for D=", D, " N=", N));

  \\ 2. L-value L(f,2) via analytic continuation
  L   = lfunmf(f);
  val = lfun(L, 2);

  \\ 3. Chowla-Selberg period Omega_D
  omega = chowla_selberg(D);

  \\ 4. alpha_2 = L(f,2)*Pi^2/Omega^4
  alpha = val * Pi^2 / omega^4;

  \\ 5. Rational recognition
  rat = bestappr(alpha, 10^(prec - 10));
  printf("D=%d  alpha_2 = %s  denom=%d\n", D, rat, denominator(rat));
  return(rat);
}

chowla_selberg(D) = {
  my(absD=abs(D), chi=kronecker(D,), w, prod=1.0);
  w = if(D==-3, 6, D==-4, 4, 2);
  forstep(a=1, absD-1, 1,
    if(gcd(a, absD)==1,
      prod *= gamma(a/absD)^(w/4 * chi(a))));
  return(prod / sqrt(2*Pi));
}

\\ Example invocations (matching Theorem 2.1):
\p 150
alpha2_cm(-4,  4)    \\ expects 1/12         (LMFDB 4.5.b.a)
alpha2_cm(-3,  12)   \\ expects 3/8          (LMFDB 12.5.b.a)
alpha2_cm(-7,  7)    \\ expects 28/3
alpha2_cm(-11, 11)   \\ expects 9/11  (first curve)
alpha2_cm(-19, 19)   \\ expects 13/57 (first curve)
alpha2_cm(-43, 43)   \\ expects 214/129
alpha2_cm(-67, 67)   \\ expects 1519/201
```

```
alpha2_cm(-163,163)  \\ expects 196216792/3 (first eigenform only)
```

Remark A.1. For $D \in \{-11, -19\}$, two distinct elliptic curves over $\mathbb{Q}$ with CM by $\mathcal{O}_{K_D}$ exist; rerun the script after substituting the alternative curve to recover both values of the twin pair. For $D = -163$, two rational eigenforms exist at level 163; only the first is covered here.

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